CS699
Lecture 1
Introduction

CS699

- Data mining is an important component of data analysis.
- Will discuss
  - Data preprocessing
  - Basic data mining algorithms
  - How to evaluate data mining models and data mining results
  - How to perform data mining using software tools

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- Prerequisites: Knowledge of R, or instructor's consent.
- R is used to illustrate algorithms.
- For the assignments and the project, you must use **primarily R** unless otherwise noted.
- R Project website:

  R: The R Project for Statistical Computing (r-project.org)
- RStudio download website:

  https://posit.co/download/rstudio-desktop/

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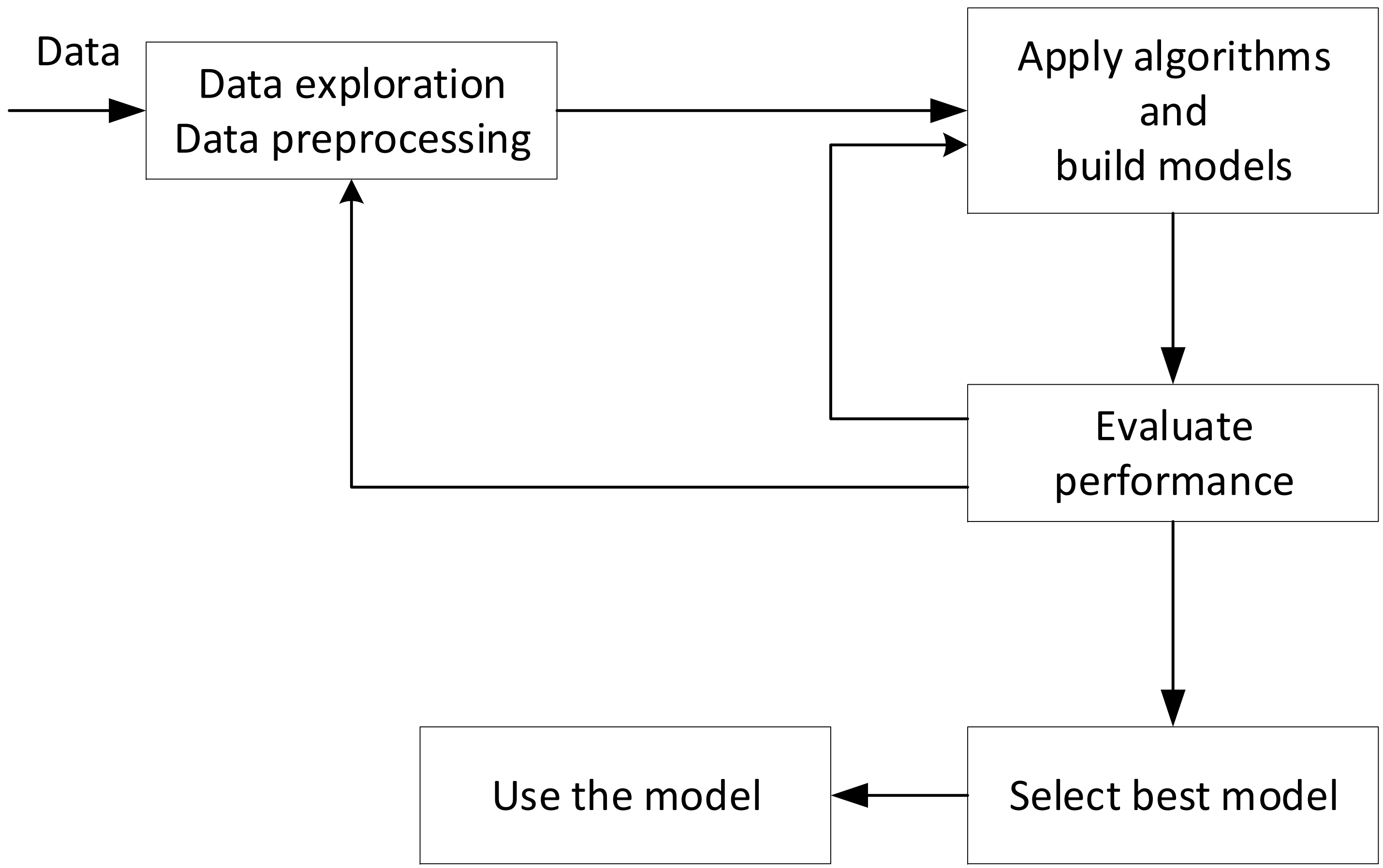
- Class project:
  - Building and testing classifier models using a real-world dataset
  - **Most of the project work should be done in R**.
  - Details will be discussed later

What Is Data Mining?

- Data mining:
  - There is no one definition
  - Has different meanings to different people and in different context
- One generic definition: knowledge discovery from data
  - Extraction of interesting (<u>non-trivial,</u> <u>implicit</u>, <u>previously unknown</u> and <u>potentially useful)</u> patterns or knowledge from huge amount of data
- Alternative names
  - Knowledge discovery (mining) in databases (KDD), predictive analytics, predictive modeling, knowledge extraction, data/pattern analysis, data archeology, data dredging, information harvesting, business intelligence, etc.

Typical Process

- Simplified, high-level process



What Kinds of Data?

- Database-oriented datasets and applications
  - Relational database, data warehouse, transactional database
- Advanced datasets and advanced applications
  - Data streams and sensor data
  - Time-series data, temporal data, sequence data (incl. bio-sequences)
  - Structure data, graphs, social networks and multi-linked data
  - Object-relational databases
  - Heterogeneous databases and legacy databases
  - Spatial data and spatiotemporal data
  - Multimedia database
  - Text databases
  - The World-Wide Web

Data Types

- Categorical (or nominal) vs. numeric data:

Categorical

OID	Age	Income	Student	Buy?
1	Young	Low	1	Y
2	Young	High	0	Y
3	Old	Low	1	N
4	Middle	Low	1	Y
5	Middle	High	1	N
6	Old	Low	0	N
7	Young	High	0	N
8	Old	High	1	Y
9	Old	High	0	Y
10	Young	Low	0	N

Numeric

OID	Age	Height	Weight
1	15	60	180
2	8	48	115
3	32	72	153
4	27	65	145
5	17	58	189
6	56	70	150
7	72	56	163
8	22	63	172
9	42	71	139
10	39	68	150

- Data types are usually tool-dependent.

Classification

- Classification and label prediction
  - Construct models (functions) based on some training examples, called training dataset.
  - Describe and distinguish classes or concepts for **future prediction**
    - E.g., classify countries based on (climate), or classify cars based on (gas mileage)
  - Predict some unknown class label (or class attribute)
- Typical methods
  - Decision trees, naïve Bayesian classification, support vector machines, neural networks, rule-based classification, pattern-based classification, logistic regression, …
- Typical applications:
  - Credit card fraud detection, direct marketing, classifying stars, diseases, web-pages, …
- Also called supervised learning

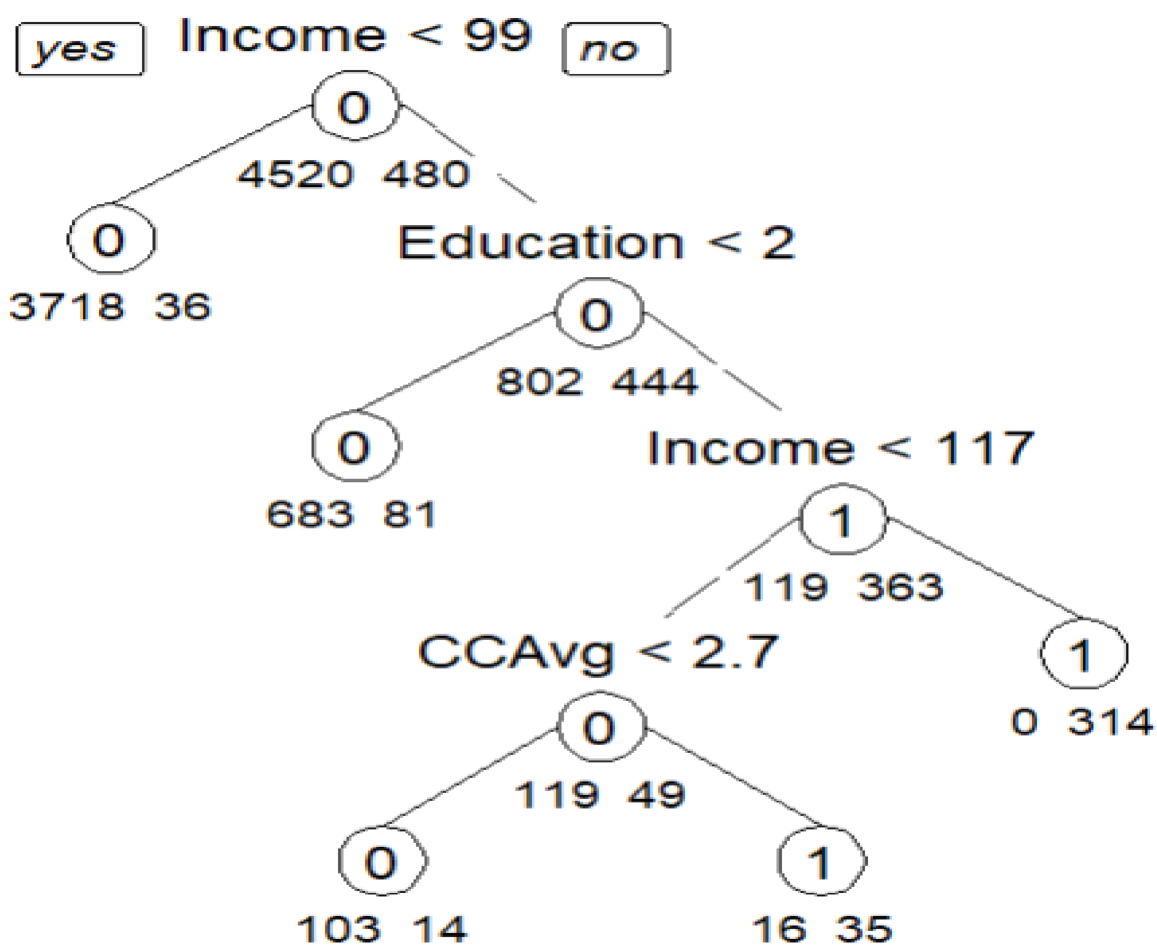
Classification Example
Universal Bank

- UniversalBank1.csv (From JMP datasets and modified)
- 6 attributes and 5000 tuples
- Predictor attributes: Age, Income, CCAvg, Education, Mortgage
- Class attribute: Personal Loan
- Goal: Build a classification model and use it to predict whether a customer will accept a personal loan
- First 5 tuples:

Personal Loan	Age	Income	CCAvg	Education	Mortgage
0	25	49	1.6	1	0
0	45	34	1.5	1	0
0	39	11	1	1	0
0	35	100	2.7	2	0
0	35	45	1	2	0

Classification Example
Universal Bank

- Decision tree built by R's *rpart* (Recursive Partitioning and Regression)



Classification vs. Numeric Prediction

- Classification:
  - Predicted (dependent) attribute is a categorical attribute.
  - Example: Predict whether a customer will buy a computer or not (yes or no, for example).
- Numeric prediction:
  - Predicted (dependent) attribute is a numeric attribute.
  - Example: Predict the weight (numeric value) of a person given the age and the height of the person.
  - Example: CPU dataset

Numeric Prediction Example

- Example: CPU dataset

  A part of the dataset

1: MYCT Numeric	2: MMIN Numeric	3: MMAX Numeric	4: CACH Numeric	5: CHMIN Numeric	6: CHMAX Numeric	7: **class** Numeric
125.0	256.0	6000.0	256.0	16.0	128.0	198.0
29.0	8000.0	32000.0	32.0	8.0	32.0	269.0
29.0	8000.0	32000.0	32.0	8.0	32.0	220.0
29.0	8000.0	32000.0	32.0	8.0	32.0	172.0
29.0	8000.0	16000.0	32.0	8.0	16.0	132.0
26.0	8000.0	32000.0	64.0	8.0	32.0	318.0
23.0	16000.0	32000.0	64.0	16.0	32.0	367.0
23.0	16000.0	32000.0	64.0	16.0	32.0	489.0
23.0	16000.0	64000.0	64.0	16.0	32.0	636.0
23.0	32000.0	64000.0	128.0	32.0	64.0	1144.0

Numeric Prediction Example

- Example: CPU dataset

  Prediction using linear regression (first 10 tuples)

MYCT	MMIN	MMAX	CACH	CHMIN	CHMAX	class	Predicted class
125	256	6000	256	16	128	198	337.18564802
29	8000	32000	32	8	32	269	311.94899965
29	8000	32000	32	8	32	220	311.94899965
29	8000	32000	32	8	32	172	311.94899965
29	8000	16000	32	8	16	132	199.08720932
26	8000	32000	64	8	32	318	332.32728062
23	16000	32000	64	16	32	367	452.35843075
23	16000	32000	64	16	32	489	452.35843075
23	16000	64000	64	16	32	636	630.64290195
23	32000	64000	128	32	64	1144	959.4871324

Association Analysis

- Frequent patterns (or frequent itemsets)
  - What items are frequently purchased together in a grocery store?
  - Mine all *frequent* itemsets and then all *strong* rules.
  - An itemset is *frequent*,

    if its support is >= predefined threshold, *minimum support*
  - A rule is written as: <left hand side> => <right hand side>
  - Example of a rule: {milk, butter} => {cheese, egg}
  - A rule is *strong*,

    if its support >= minimum support

    AND

    confidence is >= predefined threshold, *minimum confidence*

Association Analysis

- Example:

Transaction database

Customer	Items
C1	bread, chip, egg, milk
C2	beer, chip, egg, popcorn
C3	bread, chip, egg
C4	beer, bread, chip, egg, milk, popcorn
C5	beer, bread, milk
C6	beer, bread, egg
C7	bread, chip, milk
C8	bread, butter, chip, egg, milk
C9	butter, chip, egg, milk

Support Examples:

support of {bread} = 7
support of {egg, milk} = 4
support of {bread, egg, milk}
= 3

A rule R: {bread} $\Rightarrow$ {egg, milk}

Quality measures of the rule:
support(R) = support({bread, egg, milk})
= 33.3% (or 3/9) /* fraction of people who purchased {bread, milk, egg} */
confidence(R) = support(R) / support(LHS) /* conditional probability */
= 3 / 7 (or 42.9%) // among those who purchased bread, 42.9% of */
// of them also purchased {milk, egg} */

Association Analysis
Example

- Grocery purchase dataset
- A part of the dataset

Relation: grocery-ar

No.	1: cucumber Nominal	2: okra Nominal	3: squash Nominal	4: beef Nominal	5: chicken Nominal	6: orange Nominal	7: apple Nominal	8: pepper Nominal	9: tomato Nominal	10: milk Nominal
1						t	t	t		
2	t	t		t	t					
3			t				t	t	t	
4			t			t				
5	t					t				t
6		t	t	t	t	t		t		
7	t						t		t	t
8	t		t			t				
9					t		t		t	
10		t		t						

Association and Correlation Analysis

- Some association rules (mined by Apriori algorithm)

okra=t milk=t 5 ==> cucumber=t 5   <conf:(1)>

cucumber=t squash=t tomato=t 5 ==> apple=t 5   <conf:(1)>

cucumber=t squash=t apple=t 5 ==> tomato=t 5   <conf:(1)>

cucumber=t tomato=t 7 ==> apple=t 6   <conf:(0.86)>

cucumber=t okra=t 6 ==> milk=t 5   <conf:(0.83)>

cucumber=t apple=t tomato=t 6 ==> squash=t 5   <conf:(0.83)>

squash=t tomato=t 9 ==> apple=t 7   <conf:(0.78)>

squash=t apple=t 9 ==> tomato=t 7   <conf:(0.78)>

cucumber=t apple=t 8 ==> tomato=t 6   <conf:(0.75)>

milk=t 14 ==> cucumber=t 10   <conf:(0.71)>

Association and Correlation Analysis

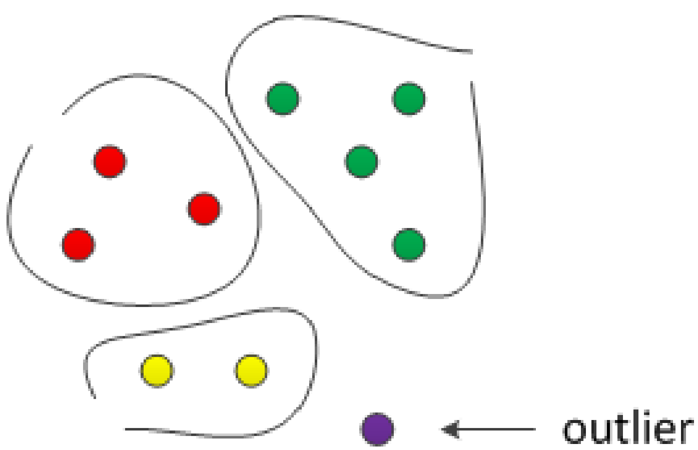
- Association, correlation vs. causality
  - Are strongly associated items also strongly correlated?
  - If two items are strongly correlated, is there a causal relationship?
- How to mine such patterns and rules efficiently in large datasets?
- Association rules can also be used for classification or clustering.

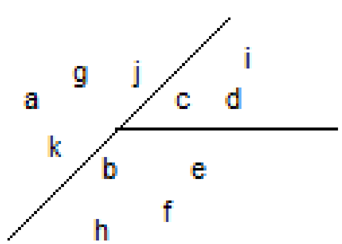
Cluster Analysis

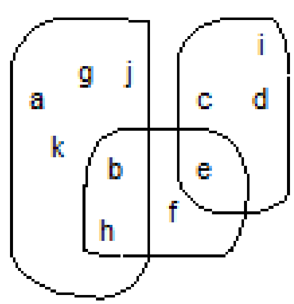
- Unsupervised learning (i.e., there is no class label)
- Group data to form new categories (i.e., clusters)
- Example: Given a customer dataset, split customers into different groups (each group is a cluster)
- Principle: Maximize intra-class similarity & minimize interclass similarity
- Many methods and applications

Cluster Analysis

- Clustering output types:







	1	2	3
a	0.8	0.1	0.1
b	0.1	0.2	0.7
c	0.6	0.3	0.1
d	0.2	0.7	0.1
e	0.1	0.1	0.8
f	0.2	0.1	0.7
g	0.7	0.2	0.1
h	0.3	0.5	0.5
i	0.2	0.6	0.2
j	0.5	0.2	0.3
k	0.1	0.2	0.7

Clustering Example
Consumer Dataset

- From JMP sample datasets; modified
- 7 attributes, 448 tuples
- Attributes: Gender, Single, School_age_children, Age, Employ_years, Salary(K), Job_Satisfaction
- Goal: Want to identify distinct groups of consumers and prepare customized promotions for different groups

Clustering Example
Consumer Dataset

- First 10 tuples

Gender	Single	School_Age_Children	Age	Employ_years	Salary(K)	Job_Satisfaction
M	No	Yes	42	14	40	Somewhat
M	No	No	49	12	48	Extremely
F	No	No	57	20	70	Extremely
M	No	Yes	52	15	50	Somewhat
M	No	Yes	44	17	41	Somewhat
M	No	No	56	32	175	Somewhat
M	Yes	No	32	3	48	Somewhat
M	No	No	28	3	100	Not_at_all
F	No	No	29	3	45	Somewhat
M	No	No	59	3	25	Extremely
M	Yes	No	41	15	150	Extremely

Clustering Example
Consumer Dataset

- Applied SPSS Two-step clustering algorithm
- Clustering result

Cluster	1	2	3	4
Size	151 (33.7%)	107 (23.9%)	101 (22.5%)	89 (19.9%)
Single	Yes (100%)	No (100%)	No (74.3%)	No (100%)
Gender	M (55.6%)	F (100%)	M (85.1%)	M (100%)
School age children	No (70.2%)	No (64.5%)	No (87.1%)	Yes (100%)
Job satisfaction	Somewhat (58.9%)	Somewhat (63.6%)	Somewhat (40.6%)	Extremely (50.6%)
Age	37.09	37.77	45.79	43.56
Employ years	6.74	8.30	11.15	11.10
Salary(K)	55.03	51.46	59.60	68.78

Outlier Analysis

- Outlier: A data object that does not comply with the general behavior of the data
- Noise or exception? — One person’s garbage could be another person’s treasure
- Methods: byproduct of clustering or regression analysis, …
- Useful in fraud detection, rare events analysis

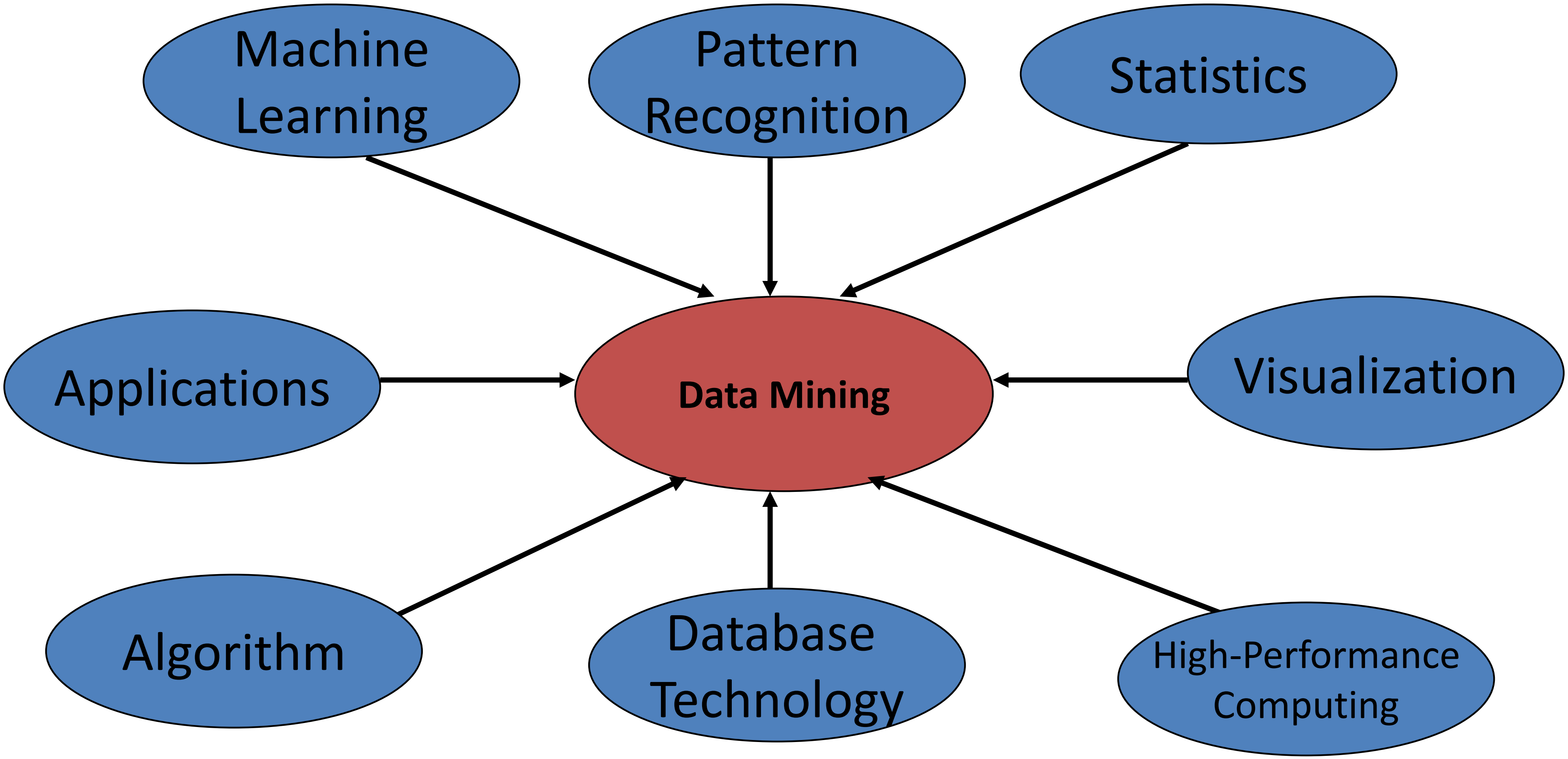
Sequential Pattern, Trend and Evolution Analysis

- Trend, time-series, and deviation analysis: e.g., regression and value prediction
- Sequential pattern mining
  - e.g., first buy digital camera, then buy large SD memory cards
- Periodicity analysis
- Biological sequence analysis

Evaluation of Knowledge

- Are all mined knowledge interesting?
  - One can mine tremendous amount of “patterns” and knowledge
  - Some may fit only certain dimension space (time, location, …)
  - Some may not be representative, may be transient, …
- A pattern is interesting if
  - easily understood
  - valid on new data or test data with some degree of certainty
  - potentially useful
  - novel
- Objective measures (e.g., support and confidence of an association rule)
- Subjective measures (e.g., expected/unexpected, actionable)

Technologies Used in Data Mining



Applications of Data Mining

- Web page analysis: from web page classification, clustering to PageRank & HITS algorithms
- Collaborative analysis & recommender systems
- Basket data analysis to targeted marketing
- Biological and medical data analysis: classification, cluster analysis (microarray data analysis), biological sequence analysis, biological network analysis
- Data mining and software engineering
- From major dedicated data mining systems/tools (e.g., SAS, MS SQL-Server Analysis Manager, Oracle Data Mining Tools) to invisible data mining

Major Issues in Data Mining

- Mining Methodology
- User Interaction
- Efficiency and Scalability
- Diversity of data types
- Data mining and society

References

- Han, J., Kamber, M., Pei, J., “Data mining: concepts and techniques,” 3rd Ed., Morgan Kaufmann, 2012
- http://www.cs.illinois.edu/~hanj/bk3/